Instant Insanity on 12 qubits

Half the qubits. No penalty terms. 44% on a real quantum simulator.

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Qiskit Fall Fest Connecticut 2026 · qBraid challenge

Four cubes. One answer.

cube 1



cube 2



cube 3



cube 4



each cube is three opposite-face pairs

the one solved tower



every side shows
all four colours
exactly once

331,776

ways to stack them

1

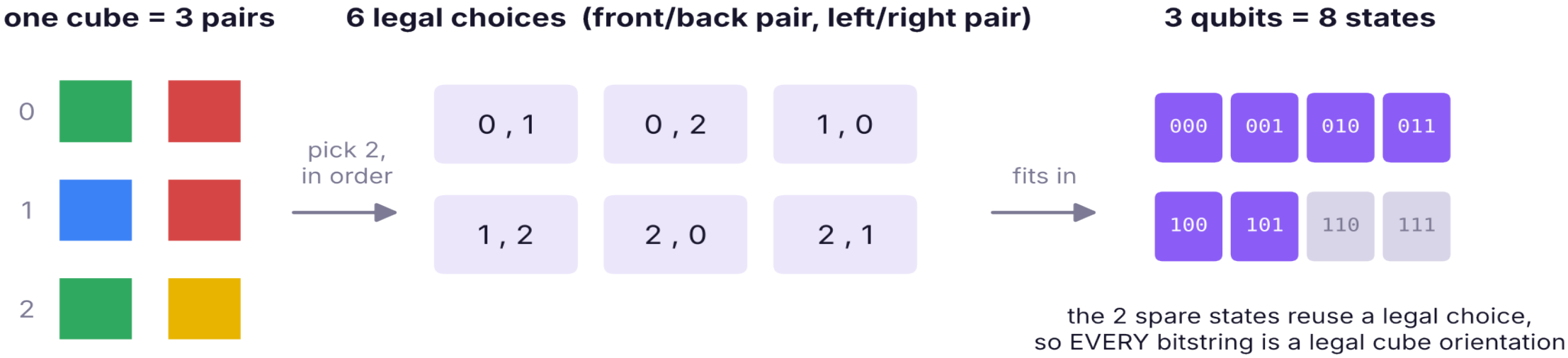
works

< 1 s

for brute force

Brute force already wins here. We say so on every slide.

The idea: every bitstring is a legal cube



24 → 12

qubits

0

penalty terms

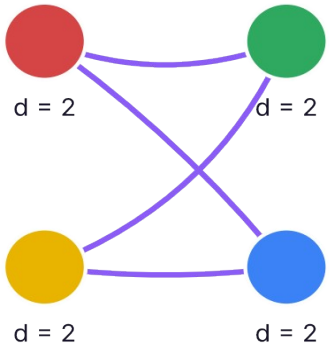
plain X

mixer, not XY

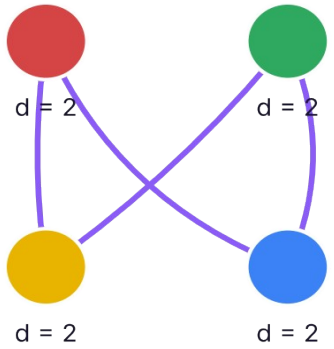
Nothing illegal to penalise, so there are no penalties.

Cost zero means solved — and we checked it four ways

front / back axis



left / right axis



one node per colour, one line per chosen pair. every colour has degree 2 → cost 0

$$\text{cost} = \Sigma\,(d_k - 2)^2$$

a sum of squares, so zero only when every colour has degree 2

1. **Hamiltonian = cost**
   on all 4,096 bitstrings, difference exactly 0
2. **Two cost implementations**
   that agree
3. **Decoder tested both ways**
   against the starter's own solves()
4. **Fast simulator = Qiskit**
   to 1e-15

We count a shot as correct only if the classical checker says so.

The circuit, and a free 2.6× saving



angles γ, β tuned classically (CVaR + layerwise growth) · 4704 CX, depth 4361 after transpiling

1216 → 460

CX per cost layer

6 → 4

longest Z string

Relabelling which states are spare made it cheaper. Same answers.

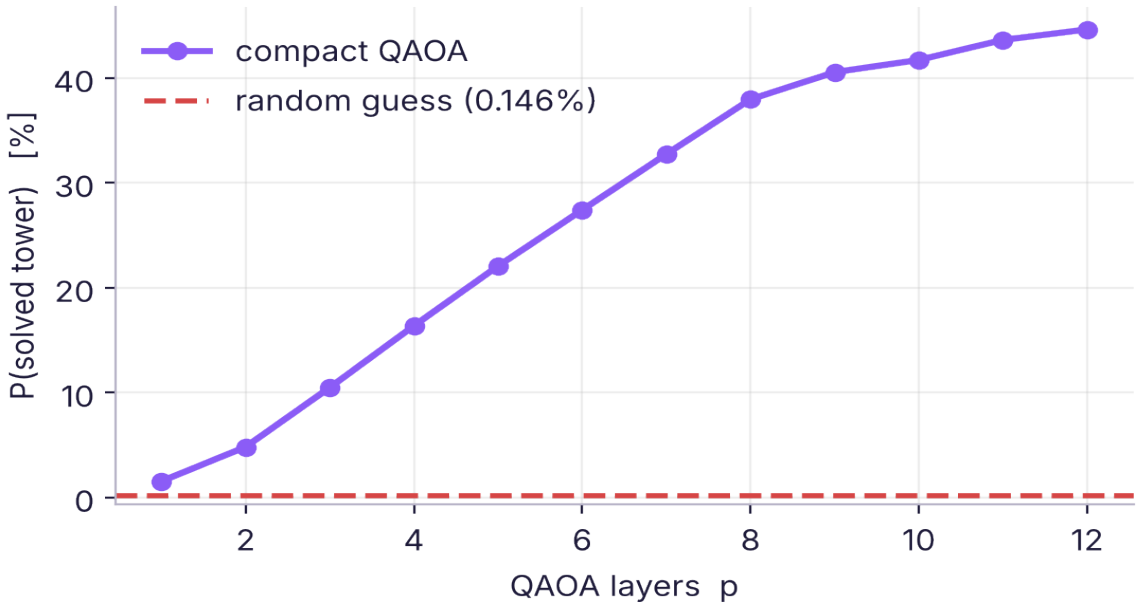
Growing the angles layer by layer

Grow layer by layer

solve p = 1, warm-start p = 2, and so on

Random restarts stall

~4%, and worse with more layers

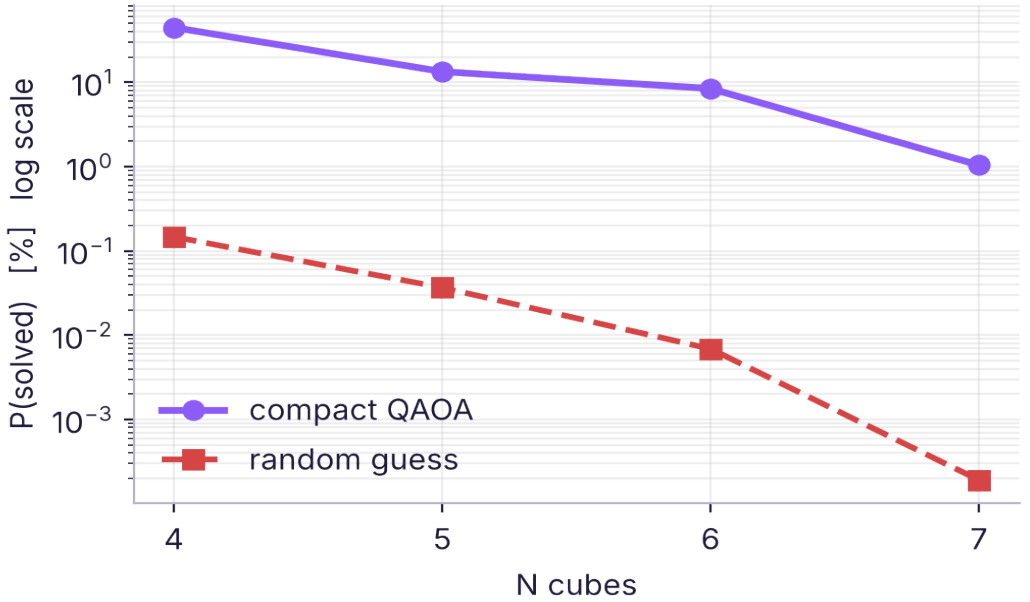


~4% → 44.6%

We also tried CVaR. An A/B showed the plain average does slightly better.

Measured on AWS SV1, and scaled to 21 qubits

N	Qubits	P(solved)	Random	Lift
4	12	44.20%	0.1465%	×302
5	15	13.37%	0.0366%	×365
6	18	8.43%	0.0069%	×1227
7	21	1.04%	0.0002%	×5457



884 of 2000 shots on SV1 = 44.20% measured

44.62% predicted — 0.4 standard errors apart

Lift grows with N. Success rate falls. We report both.

What we are not claiming

No quantum speedup.

Brute force wins at this size. We measured against random guessing inside our own encoding.

Next: share CX ladders between Z terms, and warm-start N+1 from N.

AI use and how we checked it are in the README.

